

Lab1 Part 1:

Write a paragraph about the experience and report accuracy of your success.

After reading Chapter 2 of the JM book, I understood the assignment better and I definitely felt more prepared for it. Getting the trigram dictionaries to work was fun and I enjoyed the process. I used Bayes' rule to do the probabilities which I had just learned earlier this week from my CSC 466 class so that was definitely helpful to know to get this part done. After testing, I am pretty sure I have 100% accuracy. To get the test files, I created a paragraph in English and then translated it to all of the languages and tested my language detector with that. In all, this was a fun part of the assignment!

Lab1 Part 2:

Output:

Trigram

1. Isabella can not take the whole hind-quarter.
2. Bless me!
3. Are we, probably looks much as they proceeded speaking more collectedly, with five-and-twenty thousand pounds, my first enjoyment, and whose father had judged so well able.
4. Her voice a little pleasure, sir or we should either go to her taste a great opinion of her meaning; but here, said all that to his.
5. Till you can do.

Bigram

1. Respect less watchful companion.
2. Her voice a little pleasure, sir or we should either go to her taste a great opinion of her meaning; but here, said all that to his.
3. Under one never absolutely say comfort but little spare bedchamber, hear what mr wentworth in those points offensive in authority being alone with jemima.
4. Conversation however sorry: always listening so saying if elizabeth found fresh admiration was finished my relations or happiness upon contingencies and preparing only feel quite swears by herself equal.
5. Mutual impatience all became perceptible only smiled so capable as clear estate was firmly determined solitude.

These sentences kind of look like sentences. At first, I tokenized using words which made the sentences make less sense. I switched to the sentences and added beginning of sentence and end of sentence markers to try and make the sentences make more sense.

Lab1 Part 3:

Paragraph 1

21:8 Kerioth is taken, bel is confounded by the shepherds came and pitched in dibongad. John did no more power to cast out devils? Understanding is a continual stroke, being convicted by their buttocks uncovered, yea rather, blessed both the tabernacle and all naphtali, even jericho. Howl ye; for all shall be chased as the place should kill one another any more above the house unto this mountain shall remove toward the inhabitants in his works?

Paragraph 2

11:29 Meraioth his son ministered in the lands that were ready to our children. Both young and tenderhearted, forgiving iniquity and our trespass: for more are ye saved all the craftsmen and smiths: none shall shut, it prospereth.

Paragraph 3

3:3 Resist the holy oblation foursquare, and envy, and removing to and fro like a flaming fire in them;) he was restored, and didst set thine eyes: his truth endureth to all bountifulness, which passed between those pieces. Esaias also crieth concerning israel: therefore despise him that without remedy. Terrors shall make him an ark of his master? Lament like a sharp razor, and awake to righteousness, saith thy son which shall believe on me through; so, i sit to judge this day for sodom, they washed in water face answereth to face: they toil not, my god from my youth: there will your heart to pray. Malchiram also, being reconciled, we should come together to that city.

With the help of part 2, this wasn't too different from that but extracting the verses was difficult and extracting the bible verse numbers was also hard to do as they were tokenized like "22", ":", "6". This made me have to make another helper function that would combine them so that I could have the start of the paragraph and for the random bible verse number generator at the bottom. To make the sentences seem more natural, I had a hidden end of sentence marker. This also helped vary the length of the sentences, but also made sentences too short or too long.

Lab1 Part 4:

I think you can get better and better sentences with increasing n-grams up to a certain point. When generating sentences, you don't want to have the same sentences as the training text so I would say the best sentences would come from an n-gram where $n = 5$ or 6 . These would make sentences that make sense with some randomness still. Bigrams and trigrams don't give a lot of context for sentences so they can seem jumbled up.